CLINICAL · APPROACH TO

Approach to nasal crusting

Some crusting is part of a healthy nose in a dry climate, but persistent crusting means that something has damaged the mucosa or the mucociliary clearance. The causes range from the simple, such as a dry environment, nose picking and recent surgery, through septal perforation and atrophic rhinitis, to diseases that must not be missed: granulomatosis with polyangiitis, sarcoidosis, tuberculosis, cocaine-induced destruction and malignancy. The approach separates local from systemic causes, takes biopsies and blood tests when the cause is not obvious, and relies on regular saline irrigation for almost everyone.

AT A GLANCE

- Post-surgical crusting is common and self-limiting.
- Perforation plus crusting: GPA, cocaine, TB, sarcoid, tumour.
- Bloods: ANCA, ACE, syphilis serology.
- Douching is the treatment for most.

1 CHAPTER ONE Why the nose crusts

Mucosa, airflow, clearance.

1.1 Mechanisms

Crusts form when mucus dries faster than it is cleared. This happens when the mucosa is damaged and replaced by squamous or scarred epithelium, when airflow is abnormally fast or turbulent across one area (the edge of a perforation, a wide postoperative cavity), when the cilia fail, or when the mucosa is inflamed by a granulomatous or vasculitic disease. Crusts then harbour bacteria, cause obstruction, bleeding and odour, and damage the mucosa further.

DO NOT MISS

- **Crusting with systemic symptoms** (cough, haemoptysis, arthralgia, rash, red eyes, deafness, kidney disease, weight loss): granulomatosis with polyangiitis, EGPA or sarcoidosis — ANCA, ACE, urinalysis, chest imaging, biopsy.

- **Crusting with a perforation or palatal defect in a cocaine user:** cocaine-induced midline destructive lesion; levamisole adds a vasculitis.
- **Unilateral crusting with ulceration, a mass or bleeding:** NK/T-cell lymphoma, squamous carcinoma.
- **Collapse of the nasal bridge (saddle nose):** granulomatosis with polyangiitis, relapsing polychondritis, cocaine, previous surgery.

CHAPTER TWO

2 Assessment

History, endoscopy, tests.

2.1 History

Ask how long the crusting has lasted, whether it followed surgery, and whether there is bleeding, a whistle, a bad smell (often noticed by others more than by the patient), blockage or smell loss. Ask about digital trauma, nasal sprays and drugs (including long-term decongestants and intranasal cocaine), occupational exposure to chrome compounds and industrial dusts, radiotherapy and nasogastric tubes. A full systems review looks for features of vasculitis, sarcoidosis and tuberculosis, and for chest disease, infertility and childhood infections suggesting ciliary dyskinesia or cystic fibrosis.

2.2 Examination and investigation

Examine the nose with an endoscope, before and after the crusts are softened and removed, looking for perforation, granulations, ulceration, a cobblestoned or yellow nodular mucosa (sarcoidosis), a wide cavity, and synechiae. Look at the palate, the eyes, the ears and the skin.

When the cause is not obvious, blood tests include a full blood count, renal function, inflammatory markers, **ANCA** (c-ANCA directed against proteinase-3 in granulomatosis with polyangiitis, p-ANCA against myeloperoxidase in other vasculitides), serum ACE and calcium, and, as appropriate, syphilis and tuberculosis tests. Urinalysis looks for blood and protein from renal involvement. A **biopsy** is taken from the edge of active disease, not from the crusts, when malignancy or

vasculitis is suspected; it may need repeating, because a single biopsy is often non-diagnostic. CT shows bony destruction and sinus disease.

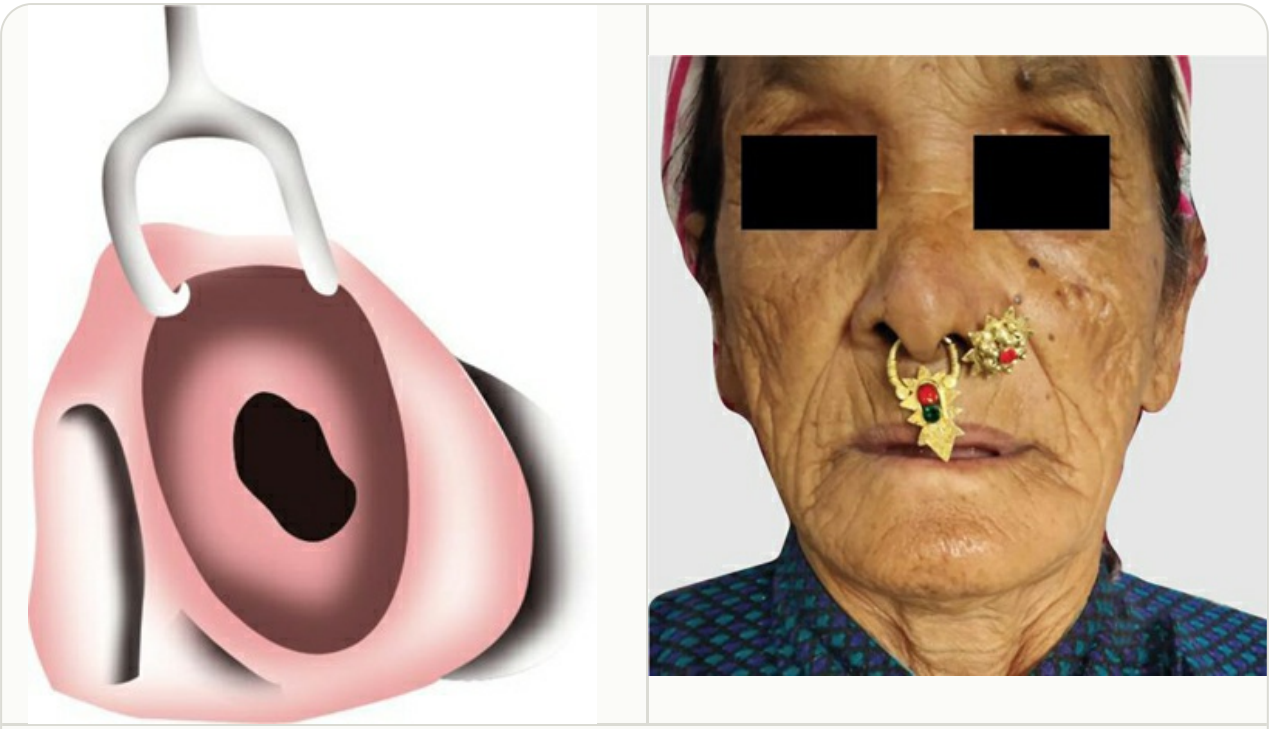


FIGURE 1 First, a septal perforation. Second, a septum deliberately perforated for an ornament.
Dhingra, figs 26.10 and 26.11

3 CHAPTER THREE Causes

Local and systemic.

CAUSES OF NASAL CRUSTING

GROUP	CAUSES	CLUES
Environmental and local	Dry air, digital trauma, decongestant overuse	Anterior septum, picking, Little's area
Surgical	After sinus surgery or turbinate reduction; empty nose syndrome	History of surgery; wide cavity
Septal perforation	Surgery, trauma, picking, cocaine, chrome workers, vasculitis	Whistling, bleeding, crusts around the edge
Atrophic rhinitis	Primary (ozaena); secondary after surgery, radiotherapy, granulomatous disease	Wide cavity, foul crusts, smell loss

GROUP	CAUSES	CLUES
Vasculitis	Granulomatosis with polyangiitis, EGPA	ANCA, systemic features, saddle nose
Granulomatous	Sarcoidosis, tuberculosis, leprosy, syphilis, rhinoscleroma	Yellow nodules, systemic features, culture
Drugs	Cocaine, levamisole	Destruction, palatal perforation
Malignancy	NK/T-cell lymphoma, squamous carcinoma	Unilateral, progressive, ulceration
Clearance	Primary ciliary dyskinesia, cystic fibrosis	Lifelong symptoms, chest disease

3.1 Atrophic rhinitis

Primary atrophic rhinitis (ozaena) occurs mainly in women in tropical and low-income settings. The mucosa and turbinates atrophy, the cavities widen, and greenish, foul-smelling crusts form, with anosmia, so the patient does not notice the odour that others do. *Klebsiella ozaenae* is often cultured. **Secondary atrophic rhinitis** follows excessive surgery, radiotherapy, trauma and granulomatous disease. **Empty nose syndrome** is a related condition after over-resection of the turbinates: the cavity is wide and clear, yet the patient feels blocked and short of breath.

3.2 Granulomatosis with polyangiitis

This necrotising vasculitis of small vessels affects the upper airway in most patients and may be limited to it. Nasal features are crusting, bleeding, obstruction, septal perforation and saddle collapse; there may also be otitis media with effusion, sensorineural hearing loss, subglottic stenosis, lung nodules and glomerulonephritis. c-ANCA supports the diagnosis, which is confirmed by biopsy and managed with rheumatology and renal teams.

4 CHAPTER FOUR Management

Irrigate, treat the cause.

4.1 General measures

Regular high-volume **saline irrigation**, several times a day, is the foundation for every cause. Lubricating ointments or oils and humidification help, and glucose-glycerine drops or warm alkaline douches soften crusts in atrophic rhinitis. Topical antibiotic ointment helps when the crusts are infected or *Staphylococcus aureus* carriage is a problem. Decongestant and steroid sprays are reviewed and stopped if they worsen crusting.

4.2 Specific treatment

Septal perforations that cause symptoms are treated with irrigation, a silicone button or surgical repair. **Atrophic rhinitis** is treated with irrigation and, for Klebsiella infection, appropriate antibiotics; surgery to narrow the cavity (Young's operation, closing the nostrils for months, or submucosal implants) is reserved for severe cases. **Vasculitis and sarcoidosis** are treated systemically with steroids and immunosuppression or biologics such as rituximab. **Cocaine-related disease** requires abstinence before any reconstruction. **Malignancy** is treated on oncological principles.

NASAL CRUSTING

1 Red flags?

→ **systemic symptoms, saddle nose or perforation**
ANCA, ACE, urinalysis, chest imaging, biopsy

→ **unilateral ulceration or mass**
biopsy and imaging

2 Clean and examine

→ **perforation**
perforation pathway

→ **wide cavity after surgery**
secondary atrophic rhinitis or empty nose

→ **foul green crusts with wide cavities**
primary atrophic rhinitis — culture

→ **anterior septum only**
dryness, picking, decongestants

3 Treat

→ saline irrigation for all

→ treat the cause

→ stop drugs that worsen crusting

→ repeat biopsy if disease progresses without a diagnosis

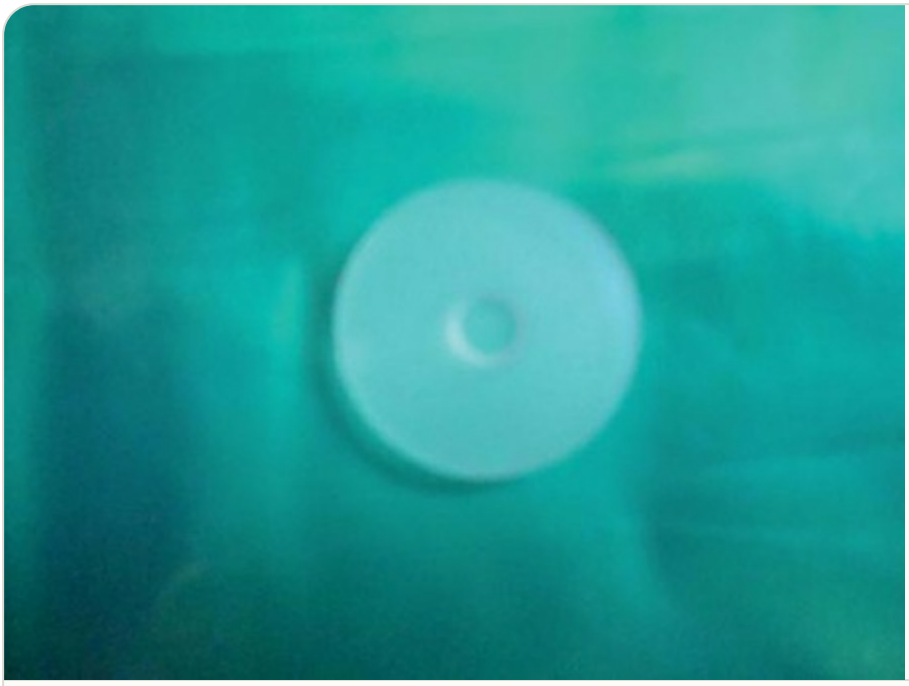


FIGURE 2 A silicone septal button closing a perforation.
Dhingra, fig. 26.12

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Local versus systemic causes** of nasal crusting.
- **The work-up:** ANCA, ACE, urinalysis, chest imaging and biopsy from the edge.
- **Atrophic rhinitis:** primary and secondary, Klebsiella ozaenae, Young's operation.
- **Granulomatosis with polyangiitis** in the upper airway.
- **Cocaine** and levamisole.

5.2 Common pitfalls

COMMON PITFALLS

- Treating vasculitis as chronic rhinosinusitis.
- Biopsying crusts instead of the active edge.
- Repairing a perforation in an active cocaine user.
- Over-resecting turbinates and creating an empty nose.

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IMAGING

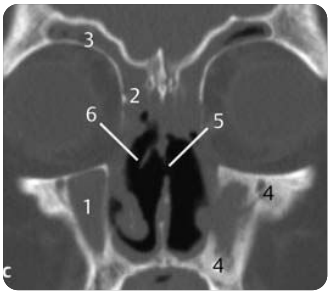
On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Granulomatosis with Polyangiitis (Wegener)

- A history of chronic sinusitis in combination with destruction of the nasal septum is highly suggestive of granulomatosis with polyangiitis (GPA). Pulmonary complaints and rapid improvement after steroid treatment may confirm this diagnosis. These therapeutic effects may also be observed in cases of allergic fungal sinusitis.
- Orbital involvement appears as a mass resulting in protrusion of the eyeball and diplopia (see also granulomatosis with polyangiitis (GPA), Orbital Involvement).

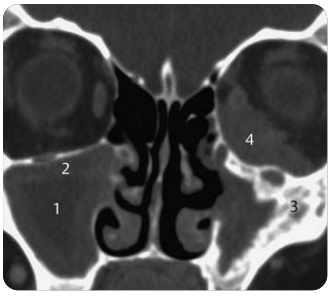


Patient with sinus pressure, nasal discharge, crusting, hyposomia, and minor epistaxis.

Dunnebier, Imaging for…

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Granulomatosis with Polyangiitis, Orbital Involvement



Patient with known granulomatosis with polyangiitis (GPA) and progressive restriction of ocular movement….